

Presidential Graduate School
DATA 210: Database Design & Analytics
Unit Test - Set A

Time: 2 Hours

Full Marks: 50

IMPORTANT EXAMINATION NOTICE

Please bring your own pens, pencils, and erasers to the examination.

*Borrowing materials from others is **strictly prohibited**. Students will not be allowed to share or request items from their peers during the session.*

Use of mobile phones is strictly prohibited. No exceptions, no excuses. **Phones out, exam over.**

Instructions:

- **Answer ALL questions.**
- Show all steps in your reasoning. For SQL questions, write the complete query; for normalization, explain each step.

Section A: Introduction to Databases **[5 Marks]**

1. Define a database and explain why databases are essential in modern software applications. Discuss the importance of databases and provide **at least three** application areas where databases play a critical role. [5]

Section B: Normalization **[10 Marks]**

2. (a) Briefly explain the concept of normalization and define **First Normal Form (1NF)**, **Second Normal Form (2NF)**, and **Third Normal Form (3NF)**. [3]
(b) Consider the following unnormalized table ENROLLMENT_UNF:

StudentID	StudentName	AdvisorID	CourseID	CourseName	Grade
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Functional Dependencies:

- StudentID → StudentName, AdvisorID
- CourseID → CourseName

- (StudentID, CourseID) is the candidate key.

The table already contains only atomic values (no multi-valued attributes).

- Is the table in 1NF? Justify your answer. [1]
- Normalize the table to **2NF**. Show the resulting tables, underline primary keys, and indicate any foreign keys. [3]
- Normalize the tables obtained in part (ii) to **3NF**. Show the final tables and explain any decompositions you make. [3]

Section C: SQL Queries

[30 Marks]

Consider the following relational schema for a company database:

Relational Schema

Employee(EmpID, EmpName, DeptID, Salary, JoinDate)

Department(DeptID, DeptName, ManagerID)

Project(ProjID, ProjName, Budget, DeptID)

Works_on(EmpID, ProjID, Hours)

Foreign Keys:

Table	Foreign Key	References
Employee	DeptID	Department(DeptID)
Department	ManagerID	Employee(EmpID)
Project	DeptID	Department(DeptID)
Works_on	EmpID	Employee(EmpID)
Works_on	ProjID	Project(ProjID)

Write SQL queries for the following:

- Retrieve the EmpID, EmpName, and Salary of all employees in the 'IT' department, sorted by Salary in descending order. Show only the first 5 records, skipping the first 2 records (use LIMIT and OFFSET). [5]
- For each department, display the department name and the average salary of its employees. Include only departments that have more than 3 employees. Use INNER JOIN and GROUP BY. [3]
 - Write a query using CROSS JOIN to list all possible combinations of department names and project names. [2]
- List the EmpName of all employees who **do not** work on any project (i.e., have no entry in Works_on). Use LEFT JOIN and check for NULL. [5]
- Combine the names of all employees and the names of all departments into a single list using UNION. Label the result column as Name and include an additional column Type indicating whether the name belongs to an 'Employee' or a 'Department'. [5]

7. Find the `EmpName` and `Salary` of employees who earn more than the average salary of all employees. Use a subquery. [5]
8. (a) Create a stored procedure named `GetProjectsByDept` that accepts a department ID and returns the names of all projects belonging to that department. Write the complete SQL code. [2.5]
(b) Create a view named `HighEarners` that displays `EmpID`, `EmpName`, and `Salary` of all employees with a salary greater than 80,000. [2.5]

Section D: NoSQL Databases

[5 Marks]

9. (a) Explain what NoSQL databases are and discuss their importance in modern data management. [2]
(b) Given a MongoDB collection named `students` with documents containing fields: `name`, `age`, `grade`, and `city`, write the following queries:
 - i. Retrieve all students with grade 'A'. [1.5]
 - ii. Find all students older than 20 and display only their `name` and `age` (projection). [1.5]